Future-Keyed Geometry Without Predictive Gain: A Linear Bridge to a Time-Reversed Reservoir

A Preprint

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August 11, 2026

registered 2026-08-10 | internal run log #3–#4 | 4 seeds, all CIs on the registered sides | replicated before publication | promoted 2026-08-11

Abstract

We construct a future-keyed representation from a frozen reservoir. A second, independent reservoir reads each text segment **backward**, so its state encodes the future of that segment; a single ridge regression — with no gradient training anywhere — maps the forward reservoir's states onto those of the time-reversed reservoir. The bridged states F(x) land far into the future-keyed region of the past–future plane: partial future-keyedness **0.51–0.55** across all four seeds, above the most future-keyed layer we have measured on a large trained transformer (0.37 — a cross-system comparison, with the caveat stated plainly below), while past-keyedness drops from 0.43–0.46 to 0.15–0.16. Nevertheless F(x) predicts the next character **0.058–0.067 bits worse** than the raw states from which it was computed, in every seed. Future-keyed geometry is obtainable from a single linear solve, and it is not what the trained readout depends on. The companion arm [x ‖ F(x)] improves on raw x by 0.026–0.035 bits in every seed: the future-keyed coordinates **add** value but cannot **replace** the past-keyed record.

Keywords reservoir computing · echo state networks · representational geometry · ridge regression · transformers · character-level language modeling · pre-registration

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§1 Introduction

This study uses a past–future faithfulness instrument. For thousands of pairs of positions in a text, it asks whether the distance between two states tracks the similarity of their *pasts* or the similarity of their *futures*, with position partialed out. Each representation lands at a point (partial_s, partial_p) on a past–future plane: partial_s measures partial faithfulness to the past (past-keyedness), partial_p measures partial faithfulness to the future (future-keyedness), each judged by paired-bootstrap confidence intervals.

Reservoir states [1] lie deep in the past-keyed region: they are very nearly a geometric record of the suffix that produced them. Layers of a large trained transformer [2] — an open-weight Llama measured earlier in this program with the same instrument conventions — sit elsewhere: its most future-keyed measured layer reaches partial_p 0.37. One plausible account holds that future-leaning geometry is an expensive product of training and is *why* trained models predict well. This study asks how expensive that geometry actually is: whether a frozen reservoir can acquire it cheaply, and, if so, whether its predictions improve.

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§2 Method

The forward reservoir is the program's reference reservoir: $N = 1024$ tanh units, $\rho = 0.8$, sparse fanin-10 recurrence. The time-reversed reservoir is an independent draw with the same parameters (seed offset +100) that reads every text segment *in reverse*. Alignment: x_t is the forward state after consuming character s_t; y_t is the backward state after consuming s_{t+1} — a summary of the text's future from t+1 onward. The bridge is a single ridge solve:

```
# one linear solve; no gradients, no training anywhere
F = argmin_F  ‖F(x_t) − y_t‖²  +  λ‖F‖²
```

Three arms are evaluated at matched readout, on 2M/500k/500k characters of text8 [3]: raw x (1024 features, the control), F(x) alone (1024 features, matched dimension — the like-for-like comparison), and [x ‖ F(x)] (2048 features — more parameters, registered and labeled as such). The faithfulness instrument is run on x and F(x) at identical positions, pairs, and bootstrap resamples (6000 positions, 20000 pairs, 1000-resample paired bootstrap), so the geometry claim is a paired comparison rather than two independent measurements.

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§3 Pre-registered predictions

- **P1** (the predicted gain): F(x) alone strictly beats raw x on logistic validation bpc — the future-keyed representation is the more usable one. The named alternative, written before the run: F(x) worse ⇒ the future-keyed representation is not more usable.
- **P1b** (low-risk consistency check): [x ‖ F(x)] beats raw x.
- **P2** (the geometry prediction): the paired-bootstrap 95% CI of Δpartial_p lies entirely above 0 *and* the CI of Δpartial_s entirely below 0.
- **P3** (a naming policy, not a claim): if *exactly one* of P1/P2 was confirmed, the dissociation itself would be the finding, pre-named as geometry without usable information.

The replication registration then fixed the claim that had to survive fresh seeds: in *every* seed, the geometry CIs on their registered sides *and* F(x) strictly worse than x. Falsifiers named in writing: any geometry break would demote the headline to an open question; any bpc break (F(x) beating x in some seed) would half-falsify it. Only 3/3 on both halves could promote the finding.

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§4 Results

P1 was not confirmed. The predicted gain did not occur and the named alternative fired: at seed 0, F(x) records 2.7178 validation bpc against raw x's 2.6512 — 0.067 bits *worse*. **P2 was confirmed decisively.** A single linear solve moved the state from (partial_s 0.4477, partial_p 0.0104) to

(0.1471, 0.5339), with CIs [+0.5071, +0.5403] on Δpartial_p and [−0.3158, −0.2848] on Δpartial_s. Exactly one of the pair was confirmed, so the pre-named dissociation became the headline result. The replication, seeds 1–3:

Table 1. Replication (internal run log #4), logistic validation bpc and paired-bootstrap 95% CIs. In every seed: CIs on the registered sides, F(x) strictly worse than x, and the concatenated arm strictly better.

seed	x	F(x)	[x ‖ F(x)]	Δpartial_p CI	Δpartial_s CI
0	2.6512	2.7178	2.6250	[+0.5071, +0.5403]	[−0.3158, −0.2848]
1	2.6497	2.7127	2.6196	[+0.5522, +0.5886]	[−0.3062, −0.2773]
2	2.6537	2.7174	2.6278	[+0.4625, +0.4975]	[−0.2862, −0.2572]
3	2.6534	2.7109	2.6184	[+0.4496, +0.4839]	[−0.2926, −0.2652]

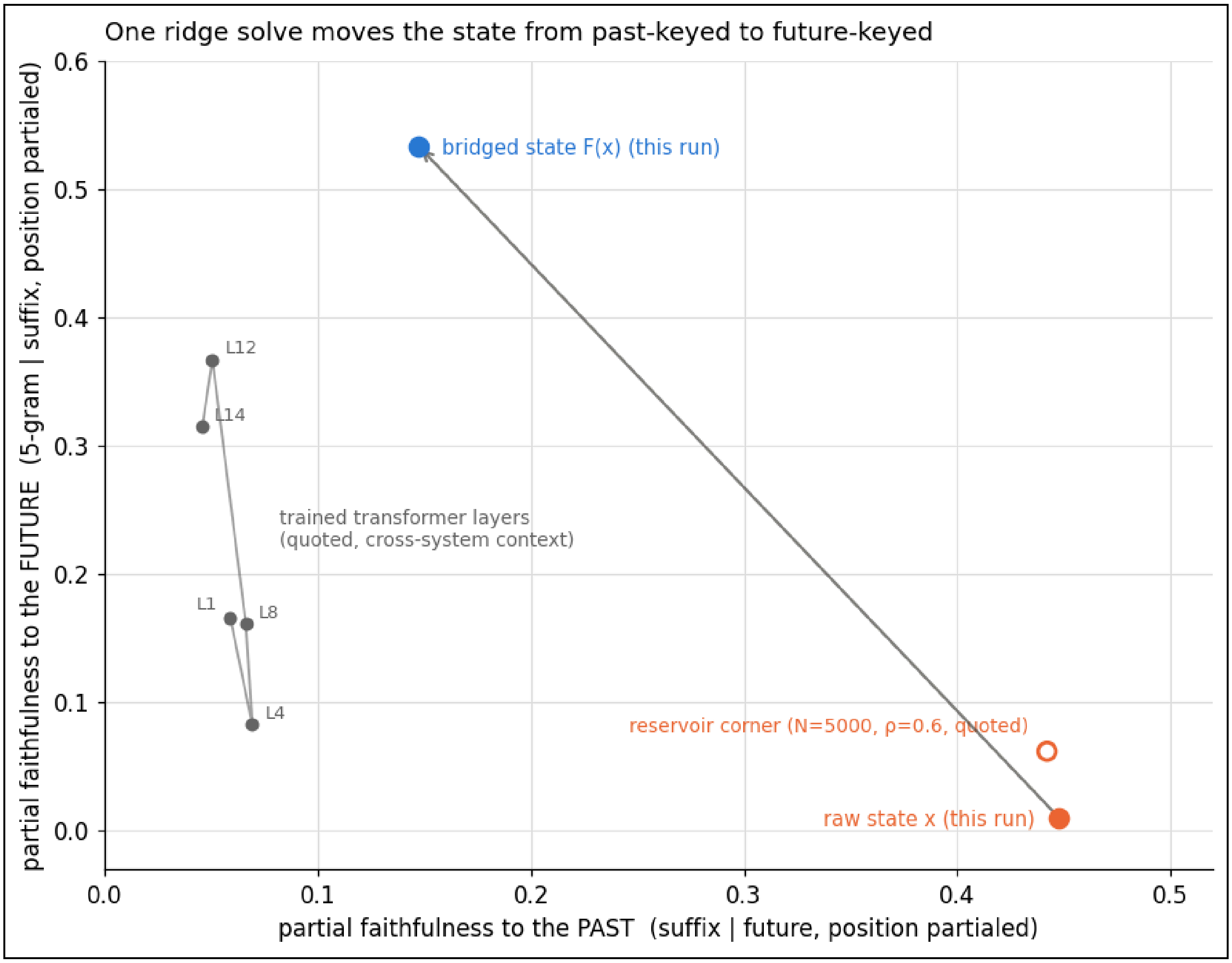

Figure 1. The bridge's move on the past–future plane, seed 0 (internal run log #3). Horizontal axis: partial faithfulness to the past (suffix, position partialed); vertical: partial faithfulness to the future (5-gram continuation, position partialed); Euclidean state distance throughout. The arrow takes raw x (0.4477, 0.0104) to F(x) (0.1471, 0.5339). Gray path: the measured Llama layers L1–L14, quoted cross-system context (most future-keyed measured layer: L12 at 0.3673). Hollow marker: the quoted N=5000, ρ=0.6 reservoir reference point.

Across all four seeds: partial_p(F(x)) spans **0.51–0.55** (0.5339 / 0.5316 / 0.5094 / 0.5525) and partial_s(F(x)) spans 0.15–0.16; the F(x) prediction penalty spans 0.058–0.067 bits; the concatenated arm's gain spans 0.026–0.035 bits. Gates passed in every seed (no readout was voided by the fit-invalidating divergence rule; bridge validation R^2 0.158–0.162 > 0; instrument sanity check 0.44–0.46).

> **The cross-system caveat, stated plainly.** The 0.37 reference point comes from measuring a Llama's layers with the same instrument conventions, but it is a different system: different substrate, different dimensionality, different training history. "More future-keyed than the most future-keyed measured Llama layer" is legitimate context for how far the bridge moves the geometry — it is *not* a controlled comparison between the reservoir and the transformer, and we rest no claim on it. The registered, controlled comparison is F(x) against its own raw x, paired cell for cell.

Two further disclosures. First, the ridge readout is blind to this entire experiment: all three arms report identical ridge validation bpc (3.1073), because F is a full-rank affine map and ridge regression at the standard λ — negligible against the feature covariance spectrum — is effectively invariant under invertible affine feature maps — every registered difference lives in the logistic readout's optimization. Second, a pilot-scale inversion disclosed at registration: at 100k training characters F(x) *beat* x by 0.068 bits; at 2M it loses by 0.067. The bridge may act as a denoising prior that helps a data-starved readout and becomes a constraint once the readout can exploit the full state — a training-set-size sweep and a reduced-rank bridge (a CCA-style solve [4]) are named, not-yet-run follow-ups.

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§5 Discussion

Position on the past–future plane is not by itself evidence of predictive skill. A frozen random reservoir plus one ridge solve occupies a more future-keyed point than any layer we have measured on a large trained model, and predicts *worse* than the states from which it was computed. Whatever future-leaning geometry contributes in trained models, the position itself is obtainable at negligible cost, so the position alone cannot account for their advantage. Representational geometry and usable predictive information are distinct quantities, and there is no fixed exchange rate between them.

The concatenated arm keeps the result from being purely negative: the future-keyed coordinates carry genuine additive value (0.026–0.035 bits in every seed); they simply cannot substitute for the past-keyed record on which the readout depends. Geometric fidelity to the future is not equivalent to the ability to predict it.

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§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
2. A. Vaswani et al., "Attention is all you need," *NeurIPS* 2017.
3. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.
4. H. Hotelling, "Relations between two sets of variates," *Biometrika* 28, 1936.

§7 Provenance

- **919b2e14** — `2026-08-10-prophet-pond`: main run, seed 0 (internal run log #3). Registered before running; P1's named alternative fired; P2 was confirmed; the pre-registered naming policy P3 produced the headline.
- **eba75ab6** — `2026-08-11-prophet-seeds-replication`: seeds 1–3 (internal run log #4). 3/3 on both registered halves; promoted by the pre-stated rule; the non-blocking concatenated-arm claim also confirmed 3/3.

All experiments registered before running, with named falsifiers; wording finalized after a disclosed 100k-character pilot run. 4 seeds (0–3); replicated before publication. Internal designation: prophet pond.
